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Google Summer of Code 2026 Project Proposal for SYMBA

Physics-Informed Models for Squared Amplitude Calculation

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- **Prerequisite Test:** `https://github.com/Ria-K912/SYMBA-GSoC-tasks`

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1 Project Introduction and Vision

1.1 Project Synopsis

The scattering cross section is one of the most fundamental observables in high-energy physics (HEP). It quantifies the probability that a given scattering process occurs and provides the direct interface between quantum field theory (QFT) predictions and experimental measurements at collider facilities. To figure out the cross section, we usually have to calculate the squared amplitude,

$$|\mathcal{M}|^2 = \mathcal{M}\,\mathcal{M}^\dagger, \tag{1}$$

followed by appropriate sums over spins and colors.

While conceptually straightforward, the symbolic computation of $|\mathcal{M}|^2$ rapidly becomes difficult as the number of external particles increases. The calculation involves intricate Lorentz contractions, Dirac algebra, color factor manipulations, propagator structures, and traces over gamma matrices. The algebraic complexity grows combinatorially with diagram multiplicity, making traditional exact symbolic frameworks such as `FeynCalc` and `MARTY` computationally expensive for multi-leg processes.

Recent advances in machine learning have opened a new direction for accelerating symbolic computations in HEP. Alnuqaydan *et al.* [1] demonstrated that sequence-to-sequence (seq2seq) transformer architectures can predict squared amplitudes directly from amplitude (or diagram) token sequences with high sequence-level accuracy-97.6% for QCD and 99% for QED while achieving inference speeds orders of magnitude faster than conventional symbolic tools. Their formulation treats amplitude squaring as a language translation problem: mapping one symbolic token sequence to another.

Although highly effective, such purely translation-based approaches do not explicitly encode the physical structure of Feynman diagrams, such as their graph topology, vertex connectivity, or propagator relations nor do they distinguish between the semantic roles of different tokens (e.g., couplings, masses, Mandelstam invariants, numerical coefficients, regulators).

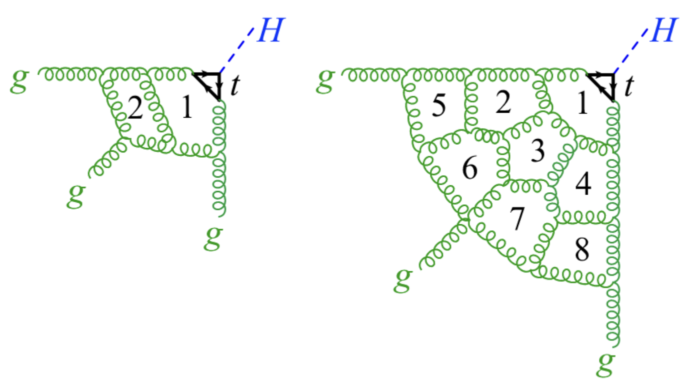

Figure 1: Sample Feynman diagrams for the process $gg \to Hg$ at two loops (left) and eight loops (right) in QCD. Adapted from Cai et al. [2].

As a result, the learned representation may rely primarily on syntactic correlations rather than physically meaningful inductive biases.

This observation motivates the development of physics-informed machine learning models that incorporate structural and semantic information intrinsic to quantum field theory.

1.2 Benefits to the Community

- **Faster and more reliable squared-amplitude computation:** Incorporating physics-informed encodings has the potential to improve predictive accuracy and generalization, particularly for complex QCD processes where purely seq2seq transformer models tend to be more data-hungry. As demonstrated by Alnuqaydan *et al.* [1], transformer-based models can achieve high sequence-level accuracy. However, embedding structural and semantic inductive biases may further reduce sample complexity while maintaining computational efficiency.
- **Reusable pipelines for physics-informed amplitude Learning:** The development of open-source preprocessing pipelines together with physics-informed models will provide a standardized benchmark and extensible foundation for the community. This enables reproducibility and facilitates future methodological comparisons.
- **Bridge between symbolic HEP and geometric deep learning:** Encoding Feynman diagrams as graphs and processing them with graph neural networks (GNNs) connects symbolic quantum field theory computations with modern geometric deep learning approaches. This aligns with recent work by Mitchell *et al.* [4] and Hou *et al.* [5] on learning directly from diagram structure, and opens the door to incorporating symmetry-aware architectures in future extensions.
- **Systematic documentation and ablation studies:** Careful documentation of architectural choices and ablation results will clarify which physics-informed components yield measurable gains. Such analysis supports ongoing work within the SYMBA project and related efforts.

1.3 Background Research

Cross sections in high-energy physics are obtained by computing the squared amplitude, averaging and summing over internal degrees of freedom (such as spin and color), and simplifying the resulting symbolic expression. Although these steps are routinely automated using computer algebra systems, the computational cost scales poorly with process complexity, especially for multi-leg QCD processes. While machine learning has achieved remarkable success in numerical high-energy physics tasks, symbolic machine learning for scattering amplitudes remains comparatively underexplored. A key insight is that mapping amplitudes to squared amplitudes can be formulated as translating one symbolic sequence into another, making sequence-to-sequence (seq2seq) transformer architectures a natural framework for the problem [1].

Alnuqaydan *et al.* [1] introduced SYMBA, a transformer-based architecture that maps

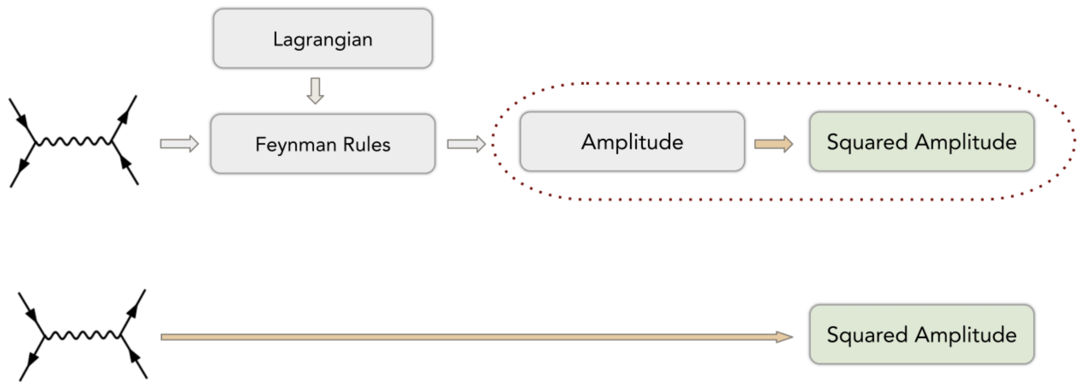

Figure 2: Amplitude to squared-amplitude and Feynman diagram to squared-amplitude workflows. Adapted from Alnuqaydan et al. [1].

amplitude (or Feynman diagram) token sequences to squared-amplitude sequences. Their model achieved sequence-level accuracies of 97.6% for QCD and 99% for QED while operating at inference speeds orders of magnitude faster than conventional symbolic tools. This work established squared-amplitude computation as a symbolic translation task amenable to transformer architectures.

Cai *et al.* [2] applied transformer architectures to compute scattering amplitude coefficients in planar $\mathcal{N} = 4$ Super Yang–Mills theory. By representing amplitudes in a structured, language-like symbolic format, they achieved accuracies exceeding 98%, demonstrating that transformers can capture highly constrained algebraic structures arising in quantum field theory.

Cheung *et al.* [3] further demonstrated that encoder-decoder transformers, combined with contrastive learning, can simplify large spinor-helicity scattering amplitude expressions. Their model reduced expressions containing hundreds of terms to compact equivalent forms such as the Parke–Taylor formula, highlighting the ability of transformers to learn nontrivial algebraic simplifications.

Mitchell *et al.* [4] explored graph-based representations of Feynman diagrams using graph attention networks (GATs). By encoding particle states as nodes and propagator relations as edges, they showed that diagram topology contains predictive information and can be leveraged to approximate matrix elements. This motivates incorporating explicit graph encodings of Feynman diagrams into symbolic learning pipelines.

Hou *et al.* [5] proposed representing high-order Feynman diagrams as computational graphs within a tensor-operation framework. Their work organizes diagram evaluation, Dyson–Schwinger equations, and renormalization procedures into structured computational graphs, facilitating integration with automatic differentiation and modern machine learning frameworks. While focused on many-electron quantum field theory, this perspective reinforces the idea that Feynman diagrams possess intrinsic graph structure that can be exploited computationally.

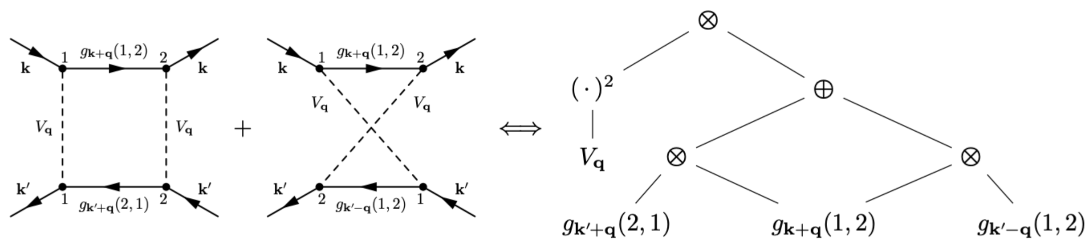

Figure 3: Illustration of the computational graph representation of two 4-point vertex function Feynman diagrams. The diagrams differ by exchange of interaction lines and are factorized into a compact computational graph consisting of propagator nodes and elementary mathematical operations (multiplication $\otimes$, addition $\oplus$, exponentiation). Adapted from Hou et al. [5].

Beyond purely symbolic prediction, neural networks have also been used to accelerate numerical amplitude evaluations in phenomenological simulations. Aylett-Bullock *et al.* [6] trained neural networks to approximate one-loop matrix elements for diphoton production and interfaced them with Monte Carlo event generators to accelerate collider simulations. Similarly, Moodie [7] demonstrated that neural-network-based approximations of one-loop amplitudes within `Sherpa` can reproduce conventional results while significantly reducing simulation time. Although these works focus on numerical rather than symbolic amplitudes, they illustrate the broader trend of applying machine learning to accelerate amplitude-related computations.

Kamienny *et al.* [8] developed an end-to-end transformer architecture for symbolic regression capable of predicting complete mathematical expressions, including constants, in a single forward pass. Vastl *et al.* [9] introduced SymFormer, further demonstrating that encoder–decoder transformers can learn structured mathematical relationships and generate exact symbolic expressions with strong generalization performance.

Finally, Lorsung *et al.* [10] proposed the Physics-Informed Token Transformer (PITT), which tokenizes partial differential equations while embedding operator structure so that the model explicitly recognizes physical operators. This motivates the introduction of physics-type token embeddings in amplitude models, where tokens corresponding to couplings, masses, Mandelstam invariants, numerical coefficients, regulators, or operators are augmented with learned embeddings reflecting their semantic roles.

Taken together, the literature establishes three complementary foundations: (i) transformer-based symbolic manipulation of amplitudes [1, 2, 3]; (ii) graph-based representations of Feynman diagrams [4, 5]; and (iii) physics-informed token embeddings and symbolic transformers for structured mathematical reasoning [8, 10, 9]. The present project builds on SYMBA by integrating diagram-graph encoding and physics-type embeddings, with systematic ablation studies to quantify the benefits of these physics-informed inductive biases.

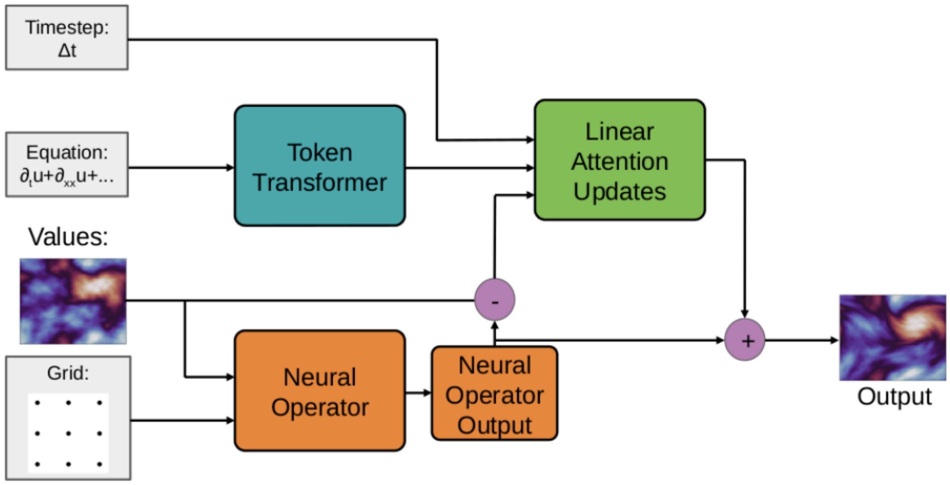

Figure 4: Architecture of the Physics-Informed Token Transformer (PITT). The model encodes governing equations using a token transformer to produce a latent embedding, which is used to guide numerical updates via linear attention blocks. The equation embedding acts as an analytically-driven correction to a data-driven neural operator. Adapted from Lorsung et al. [10].

2 Goals and Deliverables

2.1 Key Deliverables

- **Physics-informed encoding and decoding:** Design and implementation of a physics-informed architecture that integrates (i) a trainable diagram encoder capturing Feynman topology and connectivity, and (ii) physics-type token embeddings that encode the semantic roles of symbolic entities (e.g., couplings, masses, Mandelstam invariants, numerical factors, regulators). The model will fuse structural graph representations with amplitude token sequences within a unified encoder-decoder framework.
- **Exploitation of reducible structure and physics-constrained decoding:** Incorporation of known structural properties of squared amplitudes into the learning pipeline. This includes:
  * factoring out common denominators when shared across multiple terms
  * defining a fixed set of slots per process type for Mandelstam invariants, masses, and denominators, so that each term is parameterized as a coefficient with exponents over these slots
  * leveraging the known block structure of squared diagrams,
  * introducing auxiliary loss terms penalizing violations of dimensional consistency
  * implementing constraint-based decoding with iterative refinement to detect

and correct physics violations

These mechanisms aim to reduce hypothesis space, enforce physical consistency, and improve generalization.

– **Benchmarking and reproducibility:** Comprehensive benchmarking of all physics-informed variants against a vanilla transformer baseline. Evaluation will report sequence accuracy, numerical agreement, training efficiency, and inference cost. A fully reproducible pipeline will be provided, including preprocessing scripts, dataset splits, configuration files, checkpoints, and documentation.

– **Documentation and analysis:** Systematic analysis of which architectural and physics-informed choices yield measurable performance gains, under what conditions improvements occur, and why accuracy gaps arise across different process types (e.g., QED vs. QCD, low- vs. high-multiplicity). The deliverables will include recommendations for scaling strategies and future research directions within the SYMBA framework.

2.2 Prerequisite Test Overview

I have successfully completed the ML4Sci (SYMBA) 2026 prerequisite tasks (Task I and II), available at the following link: `https://github.com/Ria-K912/SYMBA-GSoC-tasks`

2.3 Technical Approach and Methodology

Theory

This project integrates physics-informed structure into sequence-to-sequence prediction for squared amplitudes. The scattering cross section is theoretically obtained by computing the squared amplitude from the amplitude. This mapping is a symbolic operation that involves Lorentz contractions, color factors, Dirac algebra, and trace manipulations, and can be formulated as translating one token sequence into another. Transformer architectures are a natural fit for this seq2seq task [1]: they predict the squared-amplitude sequence autoregressively from the amplitude (and optionally the diagram representation) using cross-entropy loss with teacher forcing during training.

Purely translation-based models, however, do not explicitly incorporate the graph structure of Feynman diagrams-namely, vertices, propagators, and external legs, nor do they distinguish the semantic role of symbols (e.g. mass versus invariant versus coupling). The proposed approach injects this structure at both the encoding and decoding stages.

- **Physics-Informed Encoding:** The Feynman diagram is parsed into a graph and processed using a graph neural network (GNN) to produce a diagram-level memory representation. In parallel, the amplitude token sequence is encoded using a transformer encoder to produce a text-level memory representation. These two memories are fused via concatenation or cross-attention-into a unified encoder memory. This allows the model to condition symbolic prediction on both diagram topology and sequential amplitude structure.
- **Physics-Informed Decoding:** Each vocabulary token is assigned a physics type (e.g. coupling, mass parameter, Mandelstam invariant, numerical coefficient, regulator, operator). A learned type embedding is added to the standard token embedding so that the decoder distinguishes between symbol roles [10]. This provides explicit semantic information and improves data efficiency and accuracy, especially for long or data-scarce sequences.
- **Exploiting Reducible Structure in Squared Amplitudes:** Squared-amplitude expressions exhibit substantial reducible structure that can be leveraged to reduce combinatorial complexity.
  * Many terms share common denominators (e.g. propagator factors); factoring these out shortens effective sequence length and reduces redundancy.
  * For a fixed process class, a structured output space can be defined with fixed slots for invariants (e.g. Mandelstam s_{ij} variables), masses, and denominators. Each term is represented as a coefficient multiplied by exponent vectors over these slots, removing permutation redundancy and reducing combinatorial growth.
  * In the vertex-edge graph representation, non-degenerate matrices arise at gluon-fermion interaction vertices. If there are k such vertices, the squared

diagram contains $\mathcal{O}(k^2)$ amplitude-conjugate pairing blocks. Encoding this vertex-block structure explicitly enables the model to respect amplitude-squaring combinatorics.

– **Physics-Constrained Learning and Decoding:** To enforce physical consistency:

  * An auxiliary loss penalizes violations of dimensional consistency (e.g. incorrect mass dimension or momentum powers).
  * During decoding, logits corresponding to physics-inconsistent tokens are masked to restrict predictions to valid symbolic continuations.
  * For non-autoregressive or chunked decoding, iterative refinement can be applied: generate an initial prediction, identify violations, mask invalid positions, and re-decode for a fixed number of iterations.

Overall Objective: By combining physics-informed encoding and decoding, structured output representations, reducible expression factoring, and explicit physics constraints, the project aims to improve accuracy, data efficiency, and scalability of squared-amplitude prediction while maintaining interpretability and alignment with the underlying structure of high-energy physics.

Task Submission

The submitted work consists of three main components:

1. **Preprocessing**

   Raw dataset entries of the form `interaction:diagram:amplitude:squared_amplitude` are loaded, parsed, and split into training, validation, and test sets. Amplitude and squared-amplitude strings are tokenized using a regex-based tokenizer that preserves physics units and composite symbolic structures as single tokens.

   Index normalization is applied per example: dummy indices are renumbered consistently within each sample to remove spurious variation, while physical indices remain unchanged to preserve semantic meaning. A shared vocabulary across amplitude and squared-amplitude tokens is constructed, and an 80–10–10 train/validation/test split is produced for downstream modeling.

2. **Physics-Informed Model**

   The model employs a dual-encoder architecture. A transformer text encoder processes the amplitude token sequence, producing a textual memory representation. In parallel, the associated Feynman diagram is parsed into a graph and processed using a graph neural network (implemented with PyTorch Geometric), yielding a diagram-level memory representation.

   These two memories are fused - either via concatenation or cross-attention with gated residual connections into a unified encoder memory.

The decoder operates autoregressively, using standard token embeddings augmented with learned physics-type embeddings that encode the semantic role of each symbol (e.g., coupling, mass, invariant, numerical coefficient). Training uses teacher forcing with cross-entropy loss. Evaluation metrics include exact sequence accuracy (full match) and token-level accuracy.

3. **Results (Physics-Informed Model)**

   Experiments were conducted on SYMBA-style datasets using an 80–10–10 split.

   - **QED.** On the QED test set (36 samples), the physics-informed model achieves **100% test sequence accuracy** and **100% token accuracy**, with stable convergence during training. The combination of diagram graph encoding and physics-type token embeddings enables exact reconstruction of the target squared-amplitude sequences.

   - **QCD.** On the QCD test set, the physics-informed model achieves **approximately 83–87% exact sequence accuracy**, with **token accuracy in the range of 94–95%**, depending on the run. QCD presents a significantly harder task than QED: sequences are longer, the vocabulary is larger, and color algebra and diagram topology introduce greater structural variability. Despite fewer training samples per structural class, the model correctly predicts a large fraction of complete sequences and most individual tokens. The graph encoder and physics-type embeddings help the model leverage diagram topology and symbolic semantics to improve performance relative to a purely sequential baseline.

The performance gap between QED and QCD reflects the increased combinatorial complexity of QCD processes, including longer symbolic expressions, richer color structure, and more diverse diagram topologies.

```
  Test Sequence Accuracy : 1.0000
  Test Token Accuracy    : 1.0000
  Test examples          : 36

Sample predictions:

  [0] TRUE : 1/36 * e ^ 4 * ( 16 * m_s ^ 2 * m_mu ^ 2 + (-8) * m_s ^ 2 * s_13 + 8 * s_14 * s_23 + (-8) * m_mu ^ 2 * s_24 + 8 * s_12
*…
      PRED : 1/36 * e ^ 4 * ( 16 * m_s ^ 2 * m_mu ^ 2 + (-8) * m_s ^ 2 * s_13 + 8 * s_14 * s_23 + (-8) * m_mu ^ 2 * s_24 + 8 * s_12
*…

  [1] TRUE : 2/81 * e ^ 4 * s_14 * s_34 * ( s_12 + 1/2 * reg_prop ) ^ (-2) + -4/81 * i * e ^ 2 * ( i * e ^ 2 * m_b ^ 2 * ( m_b ^ 2
+ …
      PRED : 2/81 * e ^ 4 * s_14 * s_34 * ( s_12 + 1/2 * reg_prop ) ^ (-2) + -4/81 * i * e ^ 2 * ( i * e ^ 2 * m_b ^ 2 * ( m_b ^ 2
+ …

  [2] TRUE : 4/81 * e ^ 4 * ( 16 * m_c ^ 2 * m_u ^ 2 + (-8) * m_c ^ 2 * s_13 + 8 * s_14 * s_23 + (-8) * m_u ^ 2 * s_24 + 8 * s_12 *
s…
      PRED : 4/81 * e ^ 4 * ( 16 * m_c ^ 2 * m_u ^ 2 + (-8) * m_c ^ 2 * s_13 + 8 * s_14 * s_23 + (-8) * m_u ^ 2 * s_24 + 8 * s_12 *
s…
```

Figure 5: Test performance of the physics-informed dual-encoder model on the QED dataset (36 test samples, 80–10–10 split). The model achieves 100% exact sequence accuracy and 100% token-level accuracy.

```
  Test Sequence Accuracy : 0.8333
  Test Token Accuracy    : 0.9488
  Test examples          : 24

Sample predictions:

  [0] TRUE : -1/6 * g ^ 4 * ( m_b ^ 2 * ( m_b ^ 2 + 1/2 * s_34 ) + s_34 * ( m_b ^ 2 + 1/2 * s_34 ) ) * ( s_12 + 1/2 * reg_prop ) ^
(-…
      PRED : -1/6 * g ^ 4 * ( m_b ^ 2 * ( m_b ^ 2 + 1/2 * s_34 ) + s_34 * ( m_b ^ 2 + 1/2 * s_34 ) ) * ( s_12 + 1/2 * reg_prop ) ^
(-…

  [1] TRUE : -1/2304 * g ^ 4 * ( s_13 * ( 128 * m_t ^ 2 + (-64) * s_13 ) + (-1024) * m_t ^ 2 * ( m_t ^ 2 + -1/2 * s_13 ) + (-224) *
m…
      PRED : -1/2304 * g ^ 4 * ( s_13 * ( 128 * m_t ^ 2 + (-64) * s_13 ) + (-1024) * m_t ^ 2 * ( m_t ^ 2 + -1/2 * s_13 ) + (-224) *
m…

  [2] TRUE : -1/16 * g ^ 4 * ( (-16) * m_c ^ 2 * m_t ^ 2 + (-8) * m_c ^ 2 * s_12 + (-8) * s_14 * s_23 + (-8) * s_13 * s_24 + (-8) *
m…
      PRED : -1/16 * g ^ 4 * ( (-16) * m_c ^ 2 * m_t ^ 2 + (-8) * m_c ^ 2 * s_12 + (-8) * s_14 * s_23 + (-8) * s_13 * s_24 + (-8) *
m…
```

Figure 6: Test performance of the physics-informed dual-encoder model on the QCD dataset (24 test samples, 80–10–10 split). The model achieves 83.33% exact sequence accuracy and 94.88% token-level accuracy. Predictions illustrate that, even when full-sequence matches are not always achieved, the majority of individual symbolic components are correctly reconstructed.

Suggested Enhancements for the Final Solution

Beyond the prerequisite tasks, the following extensions can further strengthen the physics-informed squared-amplitude pipeline:

- **Auxiliary or contrastive objectives.** Introduce auxiliary objectives that encourage structurally related inputs (e.g. amplitude-squared or diagram-squared pairs) to lie closer in embedding space than unrelated examples. This contrastive signal can improve the fused memory representation (text + graph) without modifying the primary decoding objective, potentially increasing robustness and data efficiency.
- **Two-stage decoding or constant refinement.** Structure decoding in two stages: first predict the symbolic skeleton (terms, operators, layout), then refine rational coefficients or tensor factors. This separation can reduce instability in long QCD expressions and improve numerical precision.
- **Curriculum or length-aware training.** Adopt a curriculum strategy-beginning with shorter expressions and gradually increasing maximum sequence length or use length-stratified sampling to improve convergence stability and robustness on long and structurally complex processes.
- **Symmetry-aware graph encoding.** Incorporate symmetry-aware or equivariant graph neural network layers to respect intrinsic structural and symmetry properties of Feynman diagrams. This introduces physically meaningful inductive bias and may improve generalization across process classes.
- **Systematic fusion ablations.** Compare fusion strategies (concatenation, cross-attention, gated fusion) under controlled settings while keeping all other components fixed. This clarifies how diagram and sequence representations are most

effectively integrated.

- **Common-denominator factorization.** Factor out shared propagator denominators to reduce redundancy and shorten effective sequence length. A structured representation separating numerator and denominator components may reduce combinatorial complexity.
- **Vertex–block structure encoding.** Explicitly encode the block structure arising from amplitude-conjugate amplitude pairings. This allows the model to respect the combinatorics of amplitude squaring instead of learning it implicitly from token sequences.
- **Structured output space per process class.** Define fixed slots for invariants, masses, tensor factors, coefficients, and denominators. Represent each term as exponent vectors plus a coefficient, reducing permutation redundancy and improving scalability.
- **Auxiliary dimensional-consistency loss.** Add a loss term penalizing violations of mass dimension or momentum power counting to regularize predictions toward physically valid expressions.
- **Constraint-based decoding with iterative refinement.** Mask physics-inconsistent tokens during decoding. For non-autoregressive or chunked decoding, apply iterative refinement: predict, identify violations, mask invalid positions, and re-decode for a small number of iterations.

3 Timeline and Work Plan

3.1 Application and Proposal Period (Before May 1)

During the application period, I will deepen my conceptual and technical understanding of physics-informed machine learning for symbolic amplitude computation. This includes carefully reviewing the SYMBA framework and related work on transformer-based squared-amplitude prediction, graph-based Feynman diagram representations, and structured symbolic modeling. Particular attention will be given to reducible structure in squared amplitudes, structured output representations, and constraint-aware decoding strategies.

I will validate the proposed pipeline on the existing SYMBA-style datasets, refine preprocessing components such as tokenization and diagram parsing, and conduct preliminary ablations on encoding and fusion strategies. The scope of structured-output experiments and physics-constrained decoding will be aligned with mentor guidance to ensure feasibility within a large-project timeline.

3.2 Community Bonding Period (May 1 – May 24)

This phase will focus on structured onboarding with the SYMBA project and the ML4Sci community. I will establish communication workflows with mentors and review the AMPGNN codebase and dataset formats in detail to ensure smooth integration of extensions. Development workflows will be finalized, including GitHub organization, experiment logging conventions, reproducible training scripts, and documentation templates.

A clear benchmarking protocol will be defined, covering exact sequence accuracy, token accuracy, dimensional-validity checks, and computational efficiency. The ablation plan for physics-informed encoding (diagram graph memory, fusion mechanisms) and decoding (type embeddings, constraint masking) will be finalized. Structured-output experiments,such as common-denominator handling, invariant slot representations, and vertex-block encoding will be prioritized in consultation with mentors.

3.3 Coding and Execution

Phase 1: Pre-Midterm (May 25 – July 10)

- Implement and stabilize the physics-informed dual-encoder pipeline (diagram GNN + transformer text encoder + fusion mechanism).
- Systematically evaluate fusion strategies (concatenation vs. cross-attention) and conduct controlled ablations comparing text-only, graph-only, and combined models.
- Evaluate performance on QED and QCD datasets using exact sequence accuracy, token accuracy, and training efficiency metrics.
- Prototype structured-output representations for at least one process class, including invariant slots, mass slots, and denominator handling.
- Introduce an auxiliary dimensional-consistency loss and implement an initial constraint-based decoding strategy (logit masking and/or iterative refinement).
- Prepare the midterm report summarizing results, ablations, and the roadmap for the extended phase.

Phase 2: Extended Period (July 11 – November 2)

- Expand structured-output experiments, including slot-based exponent vector representations, common-denominator factorization, and vertex-block structure encoding.
- Refine constraint-based decoding and quantitatively evaluate its impact on physical validity and exact sequence accuracy.
- Improve robustness for longer QCD sequences using curriculum learning or length-aware training strategies.

- Conduct comprehensive benchmarking comparing vanilla seq2seq, physics-informed encoding, and structured-output variants in terms of accuracy, scalability, and computational cost.
- Finalize documentation, reproducible scripts, ablation tables, and experimental analysis for project submission.

The project will conclude with well-documented code, reproducible training scripts, ablation tables and plots, and a formal project report. For a large timeline, development and evaluation will continue through November 2, ensuring thorough experimentation and documentation.

4 Academic and Technical Background

4.1 Educational Qualifications

I am currently pursuing a Dual Degree (B.E. in Computer Science and Engineering and Integrated M.Sc. in Physics) at BITS Pilani, K.K. Birla Goa Campus (CGPA: 8.84/10, graduating in 2026). My academic foundation integrates theoretical physics with core computer science, supported by coursework in Nuclear and Particle Physics, Quantum Mechanics, Machine Learning, Deep Learning, and Natural Language Processing. This interdisciplinary training enables me to approach high-energy physics problems with theoretical grounding as well as practical machine learning implementation experience.

4.2 Relevant Projects

I have engaged in a variety of machine learning research initiatives spanning academia and industry .

One of my recent works, *Beyond Accuracy: Diagnosing Reasoning Failures in Large Multimodal Models for Multimodal Physics* (Jaiswal, R., Patel, D., Singh, K., Khatoniar, R., et al.), is currently under review at ACL. This work introduces a large-scale diagnostic evaluation pipeline across five multimodal physics benchmarks (approximately 30,000 problems), performing fine-grained reasoning failure analysis beyond final-answer accuracy.

As a Research Intern at MIDAS Lab, IIIT Delhi, I built a LangChain-based debugging multi-agentic framework for repository-level natural language issues retrieval using embedding-based and monte carlo tree search.

I previously participated in Google Summer of Code 2025 with QMLHEP,ML4Sci (Project: *Quantum Kolmogorov–Arnold Networks for High Energy Physics Analysis at the LHC*). During this project, I implemented hybrid and fully quantum KAN architectures and evaluated them on Quark-Gluon and Higgs Boson datasets, gaining hands-on experience in research, and open-source development .

My additional publications include:

- Khatoniar, R., Konar, D., & Aggarwal, V. (2024). *Quantum-Enhanced Spiking Neural Networks*. In *IEEE International Conference on Quantum Computing and Engineering (QCE24)*, Montréal, Québec, Canada. Link
- Khatoniar, R., Konar, D., & Aggarwal, V. (2024). *Quantum-Enhanced Spiking Neural Networks with Temporal Encoding*. In *Quantum Techniques in Machine Learning (QTML 2024)*, Melbourne, Australia. Link
- Khatoniar, R., et al. (2024). *Quantum Convolutional Neural Network for Medical Image Classification: A Hybrid Model*. In *IEEE International Conference on Trends in Quantum Computing and Emerging Business Technologies (TQCEBT 2024)*. Link

These projects involved designing hybrid architectures, implementing training pipelines in PyTorch and Qiskit, and conducting empirical evaluations.

Additionally, as an undergraduate researcher at BITS Goa, I worked on explainability evaluation and out-of-distribution detection for AI systems, implementing counterfactual explanations, anchor-based methods, and distillation-based interpretability techniques.

I am also an active open-source contributor, having contributed to repositories such as Meetecho and SciPy. These contributions strengthened my experience with collaborative development and writing clean, maintainable code in open-source projects.

4.3 Industry Experience

I am currently a Member of Technical Staff Intern in the AI'Org at Nutanix . Here, I have worked on designing a Kubernetes-based batch inference system using custom controller and CRDs, implementing scalable file-based inference pipelines across multiple engines. I will be joining Nutanix full-time as a Member of Technical Staff in the AI'Org. This experience has strengthened my ability to build scalable, reproducible, and production-ready machine learning systems .

4.4 Motivation and Personal Drive

My research interests lie at the intersection of machine learning, natural language processing and physics. I am particularly drawn to problems where structured scientific reasoning can be modeled using modern ML architectures. Through my prior work in multimodal physics evaluation and structured reasoning analysis, I have developed a strong interest in understanding not only whether models produce correct outputs, but how they reason and where they fail. I am strongly inclined towards pursuing higher studies in machine learning for scientific applications.Contributing to this project under the guidance of experienced researchers within ML4Sci provides an invaluable opportunity to deepen my understanding of physics-informed modeling and produce reproducible research artifacts. GSoC offers not only a platform for technical contribution but also a structured environment for collaborative, open-source research : an experience that will significantly shape my academic trajectory in this field.

5 Availability and Continued Contributions

5.1 Weekly Working Hours

For the extended 350-hour GSoC project (May 25 – November 2), I will dedicate approximately 20–25 hours per week during the active coding period.

During the summer months (May–August), I will allocate:

- Weekdays: 3-4 hours each day (Monday to Friday), contributing approximately 15–16 hours.
- Weekends: 5-6 hours, contributing approximately 10-12 hours.

From August onwards, as I transition into my full-time role at Nutanix, I will maintain consistent progress by dedicating focused evening and weekend hours to the project, ensuring that milestones for the extended timeline (through November 2) are met . The extended schedule allows buffer time for rigorous experimentation, ablation studies, documentation, and refinement of output and physics-constrained components.

5.2 Communication with Mentors

I strongly value consistent and transparent communication in collaborative research projects. I will participate in regular mentor meetings to discuss progress, experimental findings, blockers, and upcoming milestones. Weekly written progress summaries will be shared, along with pre-midterm and post-midterm commits to the project repository.

I am comfortable using Slack, Discord,or email, and am also available on Mattermost for timely updates and quick clarification. Given my prior experience in ML4Sci (GSoC 2025) and open-source collaboration, I am familiar with maintaining structured documentation, experiment logs and code files.

5.3 Future Involvement Post-GSoC

My long-term academic goal is to pursue graduate studies in machine learning for scientific applications. I am enthusiastic about remaining engaged with the ML4Sci community beyond the GSoC timeline, whether through further experimentation on symbolic modeling, contributing to related projects, or extending the physics-informed framework developed during this project. I view GSoC not merely as a coding period, but as the foundation for continued research collaboration and open-source contribution .

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